

Figure 5: Adding nodes

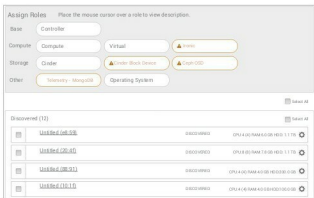


Figure 6: Assigning roles to discovered nodes



Figure 7: Roles assigned

complete the environment creation by clicking on **Create**.

Configuring the environment

Defining the networks: Define all five subnets in Fuel with the correct VLAN ID or without VLAN, based on the data centre's switch configuration.

Adding nodes: Once the environment is created, start all the other nodes with the PXE network as the boot device. Fuel will boot all nodes with the bootstrap image, which will be shown on the Web UI. Once it's discovered, click on **Add nodes**.

Figure 5 shows all the discovered nodes along with the different available roles.

Assigning roles: Assign nodes with relevant roles. In this set-up, I have assigned three nodes with *controller* roles and two nodes with *compute* roles. At least three nodes are required for controller high availability (Figure 7).

Configuring a network on discovered nodes: Figure 9 shows the configuration interface for the nodes.

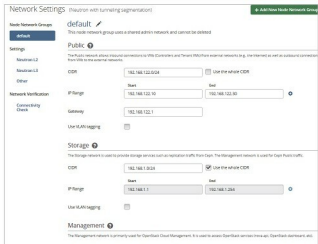


Figure 8: Defining networks

Deploying changes: Click on **Deploy changes**, which will provision the operating system using image based provision; then configure OpenStack on it with high availability for services.

Once the deployment is done, Fuel will show the IP address of the horizon dashboard, and you can log in with *admin/admin* credentials (unless changed before deployment).

Accessing nodes after deployment: Once the deployment is over, you can access all the nodes from the Fuel master. The *node.ssh* public keys are pushed to all the nodes after OS provisioning.

fuel node

This will show all the nodes, along with their node ids. To list a particular node, use the following code.

```
fuel node --node-id 23
id | status | name | cluster | ip | mac
| roles | pending_roles | online | group_id
-----+-----+-----+-----+-----+
23 | ready | controller2 | 1 | 10.20.0.13 |
52:54:00:da:ba:79 | controller | |
1 | 1
```

IP shown is the IP assigned on admin/PXE interface, and you can ssh to that node without a password.

```
ssh fuel@10.20.0.13
```

Once you are logged into any of the nodes, you will be able to see all five networks.

```
{
fuel@node-23:~$ brctl show
bridge name    bridge id    STP enabled  interfaces
```